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Distributed Swarm Trajectory Optimization for Formation Flight in Dense Environments

Lun Quan*, Longji Yin*, Chao Xu, and Fei Gao†

***Abstract*— For aerial swarms, navigation in a prescribed formation is widely practiced in various scenarios. However, the associated planning strategies typically lack the capability of avoiding obstacles in cluttered environments. To address this deficiency, we present an optimization-based method that ensures collision-free trajectory generation for formation flight. In this paper, a novel differentiable metric is proposed to quantify the overall similarity distance between formations. We then formulate this metric into an optimization framework, which achieves spatial-temporal planning using polynomial trajectories. Minimization over collision penalty is also incorporated into the framework, so that formation preservation and obstacle avoidance can be handled simultaneously. To validate the efficiency of our method, we conduct benchmark comparisons with other cutting-edge works. Integrated with an autonomous distributed aerial swarm system, the proposed method demonstrates its efficiency and robustness in real-world experiments with obstacle-rich surroundings[1]. We will release the source code for the reference of the community[2].**

I. Introduction

Autonomous aerial swarms can be employed for many systematic and cooperative tasks, such as search and rescue [1], collaborative mapping [2], and package delivery [3]. In some scenarios, it could be desired that the swarm moves according to a specified formation. For example, in [4], robots are required to form and maintain a virtual fence for animal herding tasks.

Extensive research works exist for aerial swarm navigation in formation, but none of them achieves robust formation flight in obstacle-dense environments. In practice, robots are repulsed to deviate from obstacles for safety, while formation imposes tracking targets that may oppose the obstacle avoidance. How to systematically trade off these two conflicting requirements is the key point to accomplish non-colliding formation flights. In the literature [5], consensus-based local control laws are widely used for formation maneuvering. However, the local control scheme is inherently incapable of planning over a prediction horizon, which severely undermines its practicality in complex environments. A group of optimization-based works maintain the formation by enforcing relative position constraints on each agent. Nevertheless, with this strategy, the formation would passively yield to the obstacles in dense scenarios since the surroundings could always defy the positional constraints. To summarize, a swarm trajectory planning method that can effectively manage both formation and obstacle avoidance in dense environments is lacking in the literature.

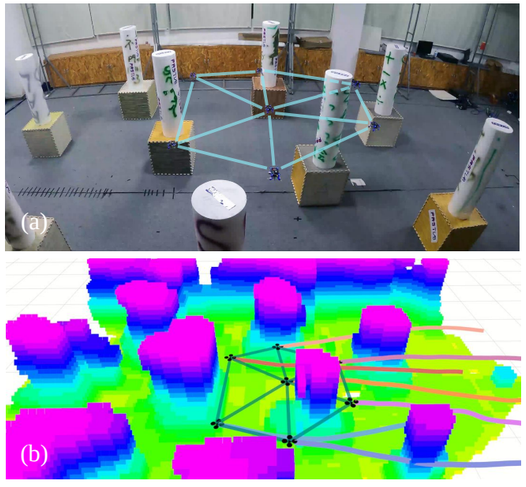


Fig. 1. A swarm of seven quadrotors in a regular hexagon formation is traversing an unknown obstacle-rich area. (a) : A snapshot of the formation flight. (b) : The visualization of the local map and executed trajectories. It should be pointed out that in this paper, the blue line segments only represent the outlines of the shape, but not the connectivity of the graph.

To bridge the gap, we propose a swarm trajectory optimization method capable of navigating swarms in formation while avoiding obstacles. We model the formation using undirected graphs and define a differentiable Laplacian-based metric that assesses the difference between formation shapes in three-dimensional workspaces. Rather than independently inspecting each agent's tracking error, our metric quantitatively evaluates the overall performance of formation maintenance, and provides greater flexibility for formation maneuvering by virtue of its invariance to translation, rotation, and scaling. To formulate the trade-offs between formation and obstacle avoidance, we design an unconstrained optimization framework that simultaneously optimizes the trajectories over feasibility cost, collision penalty, and formation similarity error. Benchmark comparisons are carried out with other state-of-the-art methods. Finally, to verify that our method is efficient and practical, extensive experiments are conducted on a real distributed aerial swarm system integrated with the

*Indicates equal contribution.

†Corresponding author: fgaoaa@zju.edu.cn.

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[1]https://www.youtube.com/watch?v=lFumt0rJci4

[2]https://github.com/ZJU-FAST-Lab/Swarm-Formation

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密集环境中的分布式群体轨迹优化以实现编队飞行

Lun Quan*，尹龙吉*，徐超，和高飞†

摘要—— 对于空中群体，在预定编队中的导航在各种场景中被广泛应用。然而，相关的规划策略通常缺乏在杂乱环境中避障的能力。为了解决这一不足，我们提出了一种基于优化的方法，确保编队飞行的无碰撞轨迹生成。本文提出了一种新颖的可微分度量，用于量化编队之间的整体相似距离。然后，我们将该度量转化为一个优化框架，利用多项式轨迹实现时空规划。在该框架中还融入了碰撞惩罚的最小化，以便同时处理编队保持和避障问题。为了验证我们方法的有效性，我们与其他前沿工作进行了基准对比。结合自主分布式空中群体系统，所提出的方法在具有丰富障碍物的实际环境中展现了其效率和鲁棒性[1]。我们将公布源代码，供社区参考[2]。

I. 引言

自主空中蜂群可以用于许多系统性和协作性任务，例如搜索与救援[1], 、协作制图 [2],以及包裹投递 [3]。在某些场景中，可能希望蜂群按照特定的队形移动。例如，在 [4],机器人需要形成并维持虚拟围栏以进行动物驱赶任务。

关于空中蜂群在队形中的导航已有大量研究工作，但没有一项能在障碍密集的环境中实现稳健的队形飞行。在实际操作中，为安全起见，机器人会被排斥偏离障碍，而队形则对跟踪目标提出了要求，这可能与避障目标相冲突。系统性地权衡这两者之间的矛盾需求，是实现非碰撞队形飞行的关键。在文献中 [5],，基于共识的局部控制律被广泛用于队形机动。然而，局部控制方案本质上无法在预测视野内进行规划，这严重削弱了其在复杂环境中的实用性。一些基于优化的方法则持续维护着通过对每个智能体施加相对位置约束来形成队形。然而，采用这种策略，队形将在密集场景中被动地让步于障碍物，因为周围环境总是可能违背位置约束。总之，文献中缺乏一种能够在密集环境中有效管理队形和避障的轨迹规划方法。

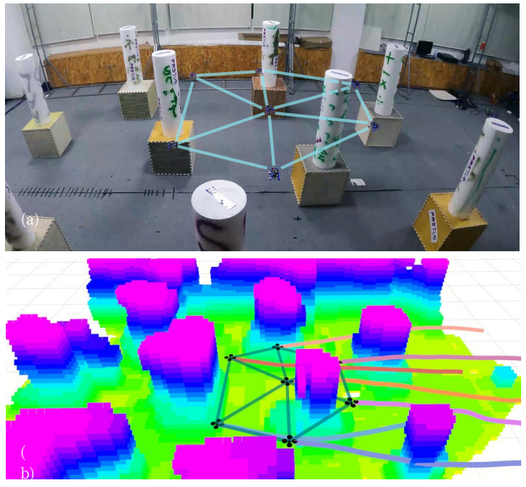


图 1. 一群七架四旋翼飞行器以正六边形队形穿越一个未知的障碍丰富区域。 (a)：队形飞行的快照。 (b)：局部地图和执行轨迹的可视化。需要指出的是，本文中蓝色线段仅代表形状的轮廓，而不代表图的连通性。

为了弥合差距，我们提出了一种群轨迹优化方法，能够在避开障碍物的同时导航成形的群体。我们使用无向图对队形进行建模，并定义了一种可微的基于拉普拉斯算子的度量，用于评估三维工作空间中队形形状之间的差异。该度量不是单独检查每个智能体的跟踪误差，而是定量评估队形维护的整体性能，并通过其对平移、旋转和缩放的不变性，为队形机动提供了更大的灵活性。为了在队形保持与避障之间权衡，我们设计了一个无约束优化框架，能够同时优化轨迹，考虑可行性成本、碰撞惩罚和队形相似性误差。我们与其他最先进的方法进行了基准对比。最后，为了验证我们的方法的高效性和实用性，在一个集成了所提出方法的真实分布式空中群系统上进行了大量实验。

*表示贡献相等。

†通讯作者：fgaoaa@zju.edu.cn。本工作得到中国国家自然科学基金资助，项目编号为62003299。所有作者均来自浙江大学工业控制技术国家重点实验室，网络科学与控制研究所，杭州 310027，中国，以及浙江大学湖州研究院，湖州 313000，中国。[1]
https://www.youtube.com/watch?v=lFumt0rJci4[2]
https://github.com/ZJU-FAST-Lab/Swarm-Formation

proposed method.

We summarize our contributions as:

1) A differentiable graph-theory-based cost function that quantifies the similarity distance between three-dimensional formations.
2) A distributed optimization framework with a joint cost function that takes formation similarity, obstacle avoidance, and dynamic feasibility into account, which makes the swarm has the ability to balance moving in formation while avoiding obstacles.
3) A series of simulations and real-world experiments with a distributed aerial swarm system that validates the efficiency and robustness of our method. The source code is released for the reference of the community[2].

II. Related works

A. Distributed Swarm Trajectory Planning

Extensive works exist for trajectory planning of distributed swarms. The concept of VO (velocity obstacle) is leveraged and generalized by Van Den Berg et al. [6]–[8] to accomplish reciprocal collision avoidance for multiple robots. However, the smoothness of the resulting trajectories cannot be guaranteed by VO-based approaches, which greatly impairs the usability on real robot systems.

In order to produce high-quality collision-free trajectories, optimization-based methods are widely introduced in the literature on distributed multicopter swarms [9]–[11]. Zhou et al. [12] incorporate Voronoi cell tessellation into a receding horizon QP scheme to prevent collision among the robots while planning. In [13], Chen et al. employ SCP to address the multi-agent planning problem in non-convex space by incrementally tightening the collision constraints. Baca et al. [14] combine MPC with a conflict resolution strategy to ensure mutual collision avoidance for outdoor swarm operations. Nevertheless, the computational load of the above optimation-based methods is large, which could hamper the applicability of the planners in highly dense scenarios.

Recently, Zhou et al. [15] present a distributed autonomous quadrotor swarm system using spatial-temporal trajectory optimization, which generates collision-free motions in dense environments merely in milliseconds. Our swarm trajectory generation scheme is based on this work.

B. Multi-robot Navigation in Formation

Various techniques have been proposed to achieve multi-robot navigation in formation, which include virtual structures [16], navigation functions [17], reactive behaviors [18], and consensus-based local control laws [19]. However, most of the existing methods only consider obstacle-free cases.

A group of works handle the formation navigation in constrained scenarios by designing feedback laws. Han et al. [20] present a formation controller based on complex-valued graph Laplacians. The formation scale is regulated by a leader to perform intended swarm maneuvering, like passing a corridor. In [21], Zhao proposes a leader-follower control law with which the formation can be affinely transformed in responding to the environmental changes. Most control approaches rely on the leader-follower scheme, where the formation parameters are only accessible to the leader.

Compared to the leader-follower scheme in [20, 21], fully decentralized strategies possess better scalability and resiliency to partial failures. In [22], Alonso-Mora et al. control a formation of drones to avoid collision by optimally rearranging the desired formation and then planning local trajectories. But since no inter-vehicle coordination exists in the distributed planners, formation maintenance is not conducted in the local planning phase. Zhou et al. [23] combine virtual structure with potential fields to produce non-colliding trajectories for formation flight. Nevertheless, planning with multiple interacting vector fields is prone to deadlocks. Besides, trajectory optimality is neglected by their method. Parys et al. [24] use DMPC to tackle the formation preservation by imposing relative position constraints on the swarm. In their framework, coordination among the agents passively breaks once the positional constraints are violated by the obstacles. In contrast, we formulate the overall formation requirement with a differentiable metric, which enables our optimizer to trade off formation and obstacle avoidance collectively and simultaneously.

III. A Differentiable Formation similarity metric

A formation of N robots is modeled by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} := \{1, 2, ..., N\}$ is the set of vertices, and $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ is the set of edges. In graph $\mathcal{G}$, the vertex i represents the ith robot with position vector $\mathbf{p}_i = [x_i, y_i, z_i] \in \mathbb{R}^3$. An edge $e_{ij} \in \mathcal{E}$ that connects vertex $i \in \mathcal{V}$ and vertex $j \in \mathcal{V}$ means the robot i and j can measure the geometric distance between each other. In our work, each robot communicates with all other robots, thus the formation graph $\mathcal{G}$ is complete. Each edge of the graph $\mathcal{G}$ is associated with a non-negative number as a weight. In this work, the weight of edge e_{ij} is given by

$$w_{ij} = \| \mathbf{p}_i - \mathbf{p}_j \|^2, \quad (i, j) \in \mathcal{E}, \tag{1}$$

where $\| \cdot \|$ denotes the Euclidean norm. Now the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ and degree matrix $\mathbf{D} \in \mathbb{R}^{N \times N}$ of the formation $\mathcal{G}$ is determined. Thus, the corresponding Laplacian matrix is given by

$$\mathbf{L} = \mathbf{D} - \mathbf{A}. \tag{2}$$

With the above matrices, the symmetric normalized laplacian matrix of graph $\mathcal{G}$ is defined as

$$\hat{\mathbf{L}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}, \tag{3}$$

where $\mathbf{I} \in \mathbb{R}^{N \times N}$ is the identity matrix.

As a graph representation matrix, Laplacian contains information about the graph structure [25]. To achieve the desired swarm formation, we propose a formation similarity distance metric as

$$f = \| \hat{\mathbf{L}} - \hat{\mathbf{L}}_{des} \|_F^2 = tr\{(\hat{\mathbf{L}} - \hat{\mathbf{L}}_{des})^T(\hat{\mathbf{L}} - \hat{\mathbf{L}}_{des})\}, \tag{4}$$

提出的方法。

我们总结了我们的贡献如下：

1) 一种基于图论的可微分代价函数，用于量化三维编队之间的相似距离。2) 一个具有联合代价函数的分布式优化框架，考虑了编队相似性、避障和动态可行性，使群体具有在保持队形的同时避开障碍的能力。3) 一系列使用分布式空中群体系统进行的仿真和实际实验，验证了我们方法的效率和鲁棒性。源代码已开源，供社区参考[2]。

二、相关工作

A. 分布式群体轨迹规划

关于分布式群体轨迹规划的研究非常丰富。VO（速度障碍）概念由Van Den Berg等人 [6]–[8] 提出并推广，用于实现多机器人之间的相互避碰。然而，基于VO的方法无法保证轨迹的平滑性，这极大地影响了其在实际机器人系统中的应用性。

为了生成高质量的无碰撞轨迹，基于优化的方法在文献中被广泛引入到分布式多旋翼无人机群 [9]–[11]。周等人 [12] 将Voronoi单元划分引入到递归视界二次规划（receding horizon QP）方案中，以在规划过程中防止机器人之间的碰撞。在 [13],中，陈等人采用SCP（逐步凸化法）解决非凸空间中的多智能体规划问题，通过逐步收紧碰撞约束。Baca等人 [14]结合模型预测控制（MPC）与冲突解决策略，确保户外群体操作中的相互避碰。然而，上述基于优化的方法计算负担较重，可能会阻碍在高度密集场景中的应用。

近年来，周等人 [15] 提出了一种基于时空轨迹优化的分布式自主四旋翼无人机群系统，该系统在密集环境中仅用毫秒级时间生成无碰撞运动。我们的群轨迹生成方案基于此项工作。

B. 多机器人编队导航

已经提出了多种技术以实现多机器人编队导航，包括虚拟结构 [16], 导航函数 [17], 反应行为 [18],以及基于共识的局部控制律 [19]。然而，现有的大多数方法仅考虑无障碍的情况。

一组工作通过设计反馈控制律来处理受限场景中的编队导航。Han 等人[20] 提出了一种基于复数图拉普拉斯矩阵的编队控制器。编队规模由领队调节，以执行预定的群体机动，例如通过走廊。在 [21],中，Zhao 提出了一种领队-跟随控制律，使得编队可以进行仿射变换在应对环境变化时，大多数控制方法依赖于领队-跟随者方案，其中编队参数仅对领队可用。

与 [20, 21],完全去中心化策略中的领航-跟随方案相比，去中心化策略具有更好的扩展性和对部分故障的鲁棒性。在 [22], Alonso-Mora等人的控制中，一组无人机通过优化重新排列期望的编队并规划局部轨迹，以避免碰撞。但由于分布式规划器中不存在车辆间的协调，局部规划阶段未进行编队维护。Zhou等人 [23]结合虚拟结构与势场，生成非碰撞的编队飞行轨迹。然而，使用多个相互作用的向量场进行规划容易陷入死锁。此外，他们的方法忽略了轨迹的最优性。Parys等人 [24] 利用DMPC通过对群体施加相对位置约束来应对编队保持问题。在他们的框架中，一旦障碍物违反了位置约束，代理之间的协调就会被动中断。相反，我们将整体编队需求用一个可微分的度量进行表达，使得我们的优化器能够同时权衡编队和避障，进行集体优化。

III. 可微分的编队相似性度量

由无向图 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ 模型化的 N 机器人编队，其中 $\mathcal{V} := \{1, 2, ..., N\}$ 是顶点集， $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ 是边集。在图 $\mathcal{G}$ 中，顶点 i 代表具有位置向量 $\mathbf{p}_i = [x_i, y_i, z_i] \in \mathbb{R}^3$ 的机器人 i^{th} 。连接顶点 $i \in \mathcal{V}$ 和顶点 $j \in \mathcal{V}$ 的边 $e_{ij} \in \mathcal{E}$ 表示机器人 i 和 j 之间可以测量几何距离。在我们的工作中，每个机器人与所有其他机器人通信，因此编队图 $\mathcal{G}$ 是完全图。图 $\mathcal{G}$ 的每条边 e_{ij} 都关联一个非负数作为权重。在本工作中，边 e_{ij} 的权重由以下公式给出：

$$w_{ij} = \| \mathbf{p}_i - \mathbf{p}_j \|^2, \quad (i, j) \in \mathcal{E}, \tag{1}$$

其中 $\| \cdot \|$ 表示欧几里得范数。现在，形成的邻接矩阵 $\mathbf{A} \in \mathbb{R}^{N \times N}$ 和度矩阵 $\mathbf{D} \in \mathbb{R}^{N \times N}$ 被确定。因此，相应的拉普拉斯矩阵为：

$$\mathbf{L} = \mathbf{D} - \mathbf{A}. \tag{2}$$

利用上述矩阵，图 $\mathcal{G}$ 的对称归一化拉普拉斯矩阵定义为

$$\hat{\mathbf{L}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}, \tag{3}$$

其中 $\mathbf{I} \in \mathbb{R}^{N \times N}$ 是单位矩阵。

作为图表示矩阵，拉普拉斯矩阵包含关于图结构 [25] 的信息。为了实现所需的群体编队，我们提出了一种编队相似距离度量为

$$f = \| \hat{\mathbf{L}} - \hat{\mathbf{L}}_{des} \|_F^2 = tr\{(\hat{\mathbf{L}} - \hat{\mathbf{L}}_{des})^T(\hat{\mathbf{L}} - \hat{\mathbf{L}}_{des})\}, \tag{4}$$

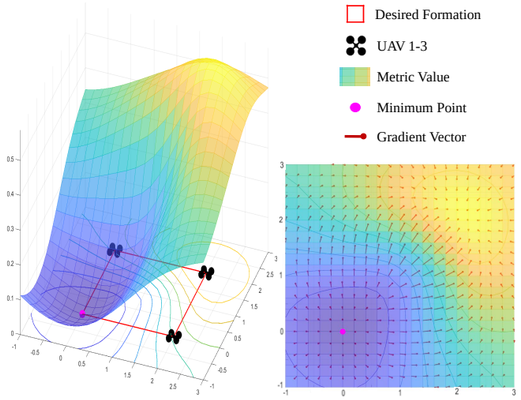


Fig. 2. Illustration of the formation similarity metric and its gradients. A square formation is desired by the swarm. The surface shows the profile of similarity metric when one UAV moves in the plane and the other three remain still. The minimum suggests the best position for the UAV to form the desired shape. Right side shows the gradients of the similarity metric.

where $tr\{\cdot\}$ denotes the trace of a matrix, $\hat{\mathbf{L}}$ is the symmetric normalized Laplacian of the current swarm formation, $\hat{\mathbf{L}}_{des}$ is the counterpart of the desired formation. Frobenius norm $\|\cdot\|_F$ is used in our distance metric. f is natively invariant to translation and rotation of the formation, since the corresponding graph is weighted by the absolute distance between robot positions. Scaling invariance is achieved by normalizing graph Laplacian with the degree matrix in (3).

Our metric is analytically differentiable with respect to the position of each robot. For robot i, we use the weights of its n adjacent edges $\{e_{i1}, e_{i2}, ..., e_{in}\}$ to form a weight vector $\mathbf{w}_i = [w_{i1}, w_{i2}, ..., w_{in}]^T$. By chain rule, the gradient of f with respect to $\mathbf{p}_i$ can be written as

$$\frac{\partial f}{\partial \mathbf{p}_i} = \frac{\partial \mathbf{w}_i}{\partial \mathbf{p}_i}\frac{\partial f}{\partial \mathbf{w}_i}. \tag{5}$$

According to our metric (4), the gradient of f with respect to each weight w_{ij} can be computed as follow

$$\frac{\partial f}{\partial w_{ij}} = tr\{(\frac{\partial f}{\partial \hat{\mathbf{L}}})^T(\frac{\partial \hat{\mathbf{L}}}{\partial w_{ij}})\}, \tag{6}$$

where

$$\frac{\partial f}{\partial \hat{\mathbf{L}}} = \frac{\partial ||\hat{\mathbf{L}} - \hat{\mathbf{L}}_{des}||_F^2}{\partial \hat{\mathbf{L}}} = 2(\hat{\mathbf{L}} - \hat{\mathbf{L}}_{des}), \tag{7}$$

$$\frac{\partial \hat{\mathbf{L}}}{\partial w_{ij}} = -\frac{\partial(\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2})}{\partial w_{ij}}. \tag{8}$$

Then the gradient $\partial f/\partial \mathbf{w}_i$ can be written as

$$\partial f/\partial \mathbf{w}_i = [\partial f/\partial w_{i1}, \partial f/\partial w_{i2}, ..., \partial f/\partial w_{in}]^T. \tag{9}$$

As for $\partial \mathbf{w}_i/\partial \mathbf{p}_i$, the Jacobian can be easily derived since the weight function (1) is differentiable. Fig.2 shows a profile of the metric and the gradient for a square formation.

IV. Spatial-Temporal Trajectory Optimization for Formation Flight

A. Trajectory Representation

In this work, we adopt the MINCO representation [26], a minimum control effort polynomial trajectory class to conduct spatial-temporal deformation of the flat-output trajectory

$$\begin{aligned}\Xi_{MINCO} =& \{p(t) : [0, T_\Sigma] \mapsto \mathbb{R}^m | \mathbf{c} = \boldsymbol{C}(\mathbf{q}, \mathbf{T}),\\ & \mathbf{q} \in \mathbb{R}^{m(M-1)}, \mathbf{T} \in \mathbb{R}^M_{>0}\},\end{aligned} \tag{10}$$

where $\mathbf{c} = (c_1^T, \cdots, c_M^T)^T$ is the polynomial coefficient, $\mathbf{q} = (q_1, \cdots, q_{M-1})$ the intermediate points, $\mathbf{T} = (T_1, \cdots, T_M)^T$ the time vector, $\boldsymbol{C}(\mathbf{q}, \mathbf{T})$ the parameter mapping constructed from Theorem 2 in [26], and $T_\Sigma = \sum_{i=1}^M T_i$ the total time.

A m-dimensional M-piece trajectory $p(t)$ is defined as

$$p(t) = p_i(t - t_{i-1}), \quad \forall t \in [t_{i-1}, t_i), \tag{11}$$

and the ith piece trajectory is represented by a $N = 5$ degree polynomial

$$p_i(t) = c_i^T \beta(t), \quad \forall t \in [0, T_i], \tag{12}$$

where $c_i \in \mathbb{R}^{6\times m}$ is the coefficient matrix, $\beta(t) = [1, t, \cdots, t^N]^T$ is the natural basis, and $T_i = t_i - t_{i-1}$ is the time allocation for the ith piece.

MINCO is uniquely determined by $(\mathbf{q}, \mathbf{T})$. And the parameter mapping $\mathbf{c} = \boldsymbol{C}(\mathbf{q}, \mathbf{T})$ converts trajectory representations $(\mathbf{c}, \mathbf{T})$ to $(\mathbf{q}, \mathbf{T})$ with linear time and space complexity, which allows any second-order continuous cost function $\boldsymbol{J}(\mathbf{c}, \mathbf{T})$ to be represented by $\widetilde{J}(\mathbf{q}, \mathbf{T})$. Hence, $\partial \widetilde{J}/\partial \mathbf{q}$ and $\partial \widetilde{J}/\partial \mathbf{T}$ can be efficiently obtained from $\partial J/\partial \mathbf{c}$ and $\partial J/\partial \mathbf{T}$, respectively.

Especially, in order to handle the time integral constraints $\psi(p(t), \cdots, p^{(3)}(t)) \preceq \mathbf{0}$, such as collision avoidance and dynamical feasibility, we transform them into finite-dimensional constraints $\psi(\hat{p}_{i,j})$ by sampling **constraint points** $\hat{p}_{i,j} = p_i((j/\kappa_i) \cdot T_i)$ on the trajectory, where κ_i is the sample number on the ith piece.

B. Optimization Problem Formulation

We formulate the trajectory generation for formation flight as an unconstrained optimization problem

$$\min_{\mathbf{c},\mathbf{T}} [J_e, J_t, J_o, J_f, J_r, J_d, J_u] \cdot \lambda, \tag{13}$$

where λ is the weight vector to trade off each cost function.

1) Control Effort J_e: The third order control input for the ith piece trajectory and its gradients are written as

$$J_e = \int_0^{T_i} \| p_i^{(3)}(t) \|^2 \, dt, \tag{14}$$

$$\frac{\partial J_e}{\partial c_i} = 2\left(\int_0^{T_i} \beta^{(3)}(t)\beta^{(3)}(t)^T dt\right) c_i, \tag{15}$$

$$\frac{\partial J_e}{\partial T_i} = c_i^T \beta^{(3)}(T_i)\beta^{(3)}(T_i)^T c_i. \tag{16}$$

2) Total Time J_t: In order to ensure the aggressiveness of the trajectory, we minimize the total time $J_t = \sum_{i=1}^M T_i$. The gradients are given by $\partial J_t/\partial \mathbf{c} = \mathbf{0}$ and $\partial J_t/\partial \mathbf{T} = \mathbf{1}$.

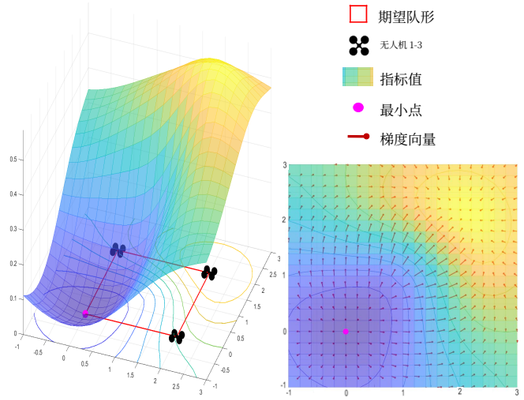


图 2. 队形相似性指标及其梯度的示意图。群体期望形成正方形队形。表面显示了当一个无人机在平面内移动而其他三架保持静止时，相似性指标的轮廓。最小值表明无人机形成期望形状的最佳位置。右侧显示了相似性指标的梯度。

其中 $tr\{\cdot\}$ 表示矩阵的迹，$\hat{}\mathbf{L}$ 是当前群体编队的对称$\hat{}$归一化拉普拉斯矩阵，$\mathbf{L}_{des}$是期望编队的对应矩阵。弗罗贝尼范数$\|\cdot\|_F$ 被用于我们的距离度量。 f 在本质上对编队的平移和旋转具有不变性，因为相应的图是由机器人位置之间的绝对距离加权的。通过在（3）中用度矩阵对图拉普拉斯矩阵进行归一化，实现了尺度不变性。

我们的指标在每个机器人位置处具有解析可微性。对于机器人 i，我们使用其n 相邻边 $\{e_{i1}, e_{i2}, ..., e_{in}\}$ 的权重来形成一个权重向量 $\mathbf{w}_i = [w_{i1}, w_{i2}, ..., w_{in}]^T$。通过链式法则，$f$ 关于$\mathbf{p}_i$ 的梯度可以写成

$$\frac{\partial f}{\partial \mathbf{p}_i} = \frac{\partial \mathbf{w}_i}{\partial \mathbf{p}_i}\frac{\partial f}{\partial \mathbf{w}_i}. \tag{5}$$

根据我们的指标（4）， f 关于每个权重 w_{ij} 的梯度可以按如下方式计算

$$\frac{\partial f}{\partial w_{ij}} = tr\{(\frac{\partial f}{\partial \hat{\mathbf{L}}})^T(\frac{\partial \hat{\mathbf{L}}}{\partial w_{ij}})\}, \tag{6}$$

其中

$$\frac{\partial f}{\partial \hat{\mathbf{L}}} = \frac{\partial ||\hat{\mathbf{L}} - \hat{\mathbf{L}}_{des}||_F^2}{\partial \hat{\mathbf{L}}} = 2(\hat{\mathbf{L}} - \hat{\mathbf{L}}_{des}), \tag{7}$$

$$\frac{\partial \hat{\mathbf{L}}}{\partial w_{ij}} = -\frac{\partial(\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2})}{\partial w_{ij}}. \tag{8}$$

然后，梯度 $\partial f/\partial \mathbf{w}_i$ 可以写成

$$\partial f/\partial \mathbf{w}_i = [\partial f/\partial w_{i1}, \partial f/\partial w_{i2}, ..., \partial f/\partial w_{in}]^T. \tag{9}$$

至于 $\partial \mathbf{w}_i/\partial \mathbf{p}_i$，雅可比矩阵可以轻松推导，因为权重函数（1）是可微的。图2显示了一个正方形队形的度量和梯度的轮廓。

A. 轨迹表示

在本工作中，我们采用 MINCO 表示 [26]，一种最小控制努力多项式轨迹类别，用于进行平坦输出轨迹的时空变形。

$$\begin{aligned}\Xi_{MINCO} =& \{p(t) : [0, T_\Sigma] \mapsto \mathbb{R}^m | \mathbf{c} = \boldsymbol{C}(\mathbf{q}, \mathbf{T}),\\ & \mathbf{q} \in \mathbb{R}^{m(M-1)}, \mathbf{T} \in \mathbb{R}^M_{>0}\},\end{aligned} \tag{10}$$

其中 $\mathbf{c} = (c_1^T, \cdots, c_M^T)^T$ 是多项式系数，$\mathbf{q} = (q_1, \cdots, q_{M-1})$ 是中间点，$\mathbf{T} = (T_1, \cdots, T_M)^T$ 是时间向量，$\boldsymbol{C}(\mathbf{q},\mathbf{T})$ 是由定理 2 构建的参数映射，[26]，和 $T_\Sigma = \sum_{i=1}^M T_i$ 是总时间。

一个 m 维 M 段轨迹 $p(t)$ 被定义为

$$p(t) = p_i(t - t_{i-1}), \quad \forall t \in [t_{i-1}, t_i), \tag{11}$$

并且该 i^{th} 段轨迹由一个 $N = 5$ 次多项式表示

$$p_i(t) = c_i^T \beta(t), \quad \forall t \in [0, T_i], \tag{12}$$

其中 $c_i \in \mathbb{R}^{6\times m}$ 是系数矩阵， $\beta(t) = [1, t, \cdots, t^N]^T$ 是自然基， $T_i = t_i - t_{i-1}$ 是该 i^{th} 段的时间分配。

MINCO 由 $(\mathbf{q},\mathbf{T})$ 唯一确定。参数映射 $\mathbf{c} = \boldsymbol{C}(\mathbf{q},\mathbf{T})$ 将轨迹表示 $(\mathbf{c},\mathbf{T})$ 转换为 $(\mathbf{q},\mathbf{T})$，具有线性时间和空间复杂度，这允许任何二阶连续的代价函数 $\boldsymbol{J}(\mathbf{c},\mathbf{T})$ 被 $\widetilde{J}(\mathbf{q},\mathbf{T})$ 表示为 $\boldsymbol{J}(\mathbf{q},\mathbf{T})$。因此， $\partial J/\partial \mathbf{q}$ 和 $\partial J/\partial \mathbf{T}$ 可以分别高效地从 $\partial J/\partial \mathbf{c}$ 和 $\partial J/\partial \mathbf{T}$ 获取。

尤其，为了处理时间积分约束 $\psi(p(t) \cdots, p^{(3)}(t)) \preceq \mathbf{0}$，例如碰撞避免和动力学可行性，我们通过在轨迹上采样 **约束点** $\hat{}p_{i,j} = p_i((j/\kappa_i) \cdot T_i)$ 将它们转化为有限维约束 $\psi(\hat{}p_{i,j})$，其中 κ_i 是在 i^{th} 段上的采样点数。

B. 优化问题的建模

我们将编队飞行的轨迹生成问题表述为一个无约束优化问题

$$\min_{\mathbf{c},\mathbf{T}} [J_e, J_t, J_o, J_f, J_r, J_d, J_u] \cdot \lambda, \tag{13}$$

其中 λ 是用于权衡各个代价函数的权重向量。

1) 控制努力 J_e: 用于 i^{th} 段轨迹的三阶控制输入及其梯度表示为

$$J_e = \int_0^{T_i} \| p_i^{(3)}(t) \|^2 \, dt, \tag{14}$$

$$\frac{\partial J_e}{\partial c_i} = 2\left(\int_0^{T_i} \beta^{(3)}(t)\beta^{(3)}(t)^T dt\right) c_i, \tag{15}$$

$$\frac{\partial J_e}{\partial T_i} = c_i^T \beta^{(3)}(T_i)\beta^{(3)}(T_i)^T c_i. \tag{16}$$

2) 总时间 J_t: 为了确保轨迹的激进性，我们最小化总时间 $J_t = \sum_{i=1}^M T_i$。梯度由 $\partial J_t/\partial \mathbf{c} = \mathbf{0}$ 和 $\partial J_t/\partial \mathbf{T} = \mathbf{1}$ 给出。

3) Obstacle Avoidance J_o: Inspired by [27], obstacle avoidance penalty J_o is computed using Euclidean Signed Distance Field (ESDF). The constraint points which are close to the obstacles are selected by

$$\psi_o(\hat{p}_{i,j}) = \begin{cases} d_{thr} - d(\hat{p}_{i,j}), & \text{if} \quad d(\hat{p}_{i,j}) < d_{thr}, \\ 0, & \text{if} \quad d(\hat{p}_{i,j}) \geq d_{thr}, \end{cases} \tag{17}$$

where d_{thr} is the safety threshold and $d(\hat{p}_{i,j})$ is the distance between the considered point and the closest obstacle around it. Then the obstacle avoidance penalty is obtained by computing the weighted sum of sampled constraint function:

$$J_o = \frac{T_i}{\kappa_i} \sum_{j=0}^{\kappa_i} \bar{\omega}_j \max\{\psi_o(\hat{p}_{i,j}), 0\}^3, \tag{18}$$

where $(\bar{\omega}_0, \bar{\omega}_1, \cdots, \bar{\omega}_{\kappa_i - 1}, \bar{\omega}_{\kappa_i}) = (1/2, 1, \cdots, 1, 1/2)$ are the orthogonal coefficients following the trapezoidal rule [28].

The gradients of J_o w.r.t c_i and T_i are detailed as

$$\frac{\partial J_o}{\partial c_i} = \frac{\partial J_o}{\partial \psi_o} \frac{\partial \psi_o}{\partial c_i}, \tag{19}$$

$$\frac{\partial J_o}{\partial T_i} = \frac{J_o}{T_i} + \frac{\partial J_o}{\partial \psi_o} \frac{\partial \psi_o}{\partial t} \frac{\partial t}{\partial T_i}, \tag{20}$$

$$\frac{\partial t}{\partial T_i} = \frac{j}{\kappa_i}, \qquad t = \frac{j}{\kappa_i} T_i, \tag{21}$$

where t is the relative time on the piece. For the case that $d(\hat{p}_{i,j}) < d_{thr}$, the gradients are given by

$$\frac{\partial \psi_o}{\partial c_i} = -\beta(t) \nabla d^T, \qquad \frac{\partial \psi_o}{\partial t} = -\nabla d^T \dot{p}(t), \tag{22}$$

where ∇d is the gradient of ESDF in $\hat{p}_{i,j}$. Otherwise, the gradients become $\partial \psi_o / \partial c_i = \mathbf{0}$, $\partial \psi_o / \partial t = 0$.

4) Swarm Formation Similarity J_f: In Sec.III, we design a differentiable metric to quantify the similarity distance between swarm formations. In optimization, the similarity error between the current formation and the desired formation is measured by $\psi_f = f(p(t), \bigcup_\Phi p_\phi(\tau))$, where $f(\cdot)$ is detailed in (4) and Φ represents the collection of other agents.

Since J_f involves the trajectories of other agents, we need to deal with both the relative time $t = jT_i/\kappa_i$ of the own trajectory and the global timestamp $\tau = T_1 + ... + T_{i-1} + jT_i/\kappa_i$ of others' trajectories. J_f considers the preceding time T_l for any $1 \leq l \leq i$ and is formulated as

$$J_f = \frac{T_i}{\kappa_i} \sum_{j=0}^{\kappa_i} \bar{\omega}_j \max\{\psi_f(p(t), \bigcup_\Phi p_\phi(\tau)), 0\}^3. \tag{23}$$

The gradients of J_f w.r.t c_i and T_l are computed as

$$\frac{\partial J_f}{\partial c_i} = \frac{\partial J_f}{\partial \psi_f} \frac{\partial \psi_f}{\partial c_i}, \tag{24}$$

$$\frac{\partial J_f}{\partial T_l} = \frac{J_f}{T_l} + \frac{\partial J_f}{\partial \psi_f} \frac{\partial \psi_f}{\partial T_l}. \tag{25}$$

To derive $\partial \psi_f / \partial T_l$, ψ_f need to be differentiated by t and τ:

$$\frac{\partial \psi_f}{\partial T_l} = \frac{\partial \psi_f}{\partial t} \frac{\partial t}{\partial T_l} + \frac{\partial \psi_f}{\partial \tau} \frac{\partial \tau}{\partial T_l}, \tag{26}$$

$$\frac{\partial t}{\partial T_l} = \begin{cases} \frac{j}{\kappa_i}, & l = i, \\ 0, & l < i, \end{cases} \qquad \frac{\partial \tau}{\partial T_l} = \begin{cases} \frac{j}{\kappa_i}, & l = i, \\ 1, & l < i. \end{cases} \tag{27}$$

The gradients of ψ_f w.r.t c_i, t and τ are given by

$$\frac{\partial \psi_f}{\partial c_i} = \frac{\partial \psi_f}{\partial p(t)} \frac{\partial p(t)}{\partial c_i}, \tag{28}$$

$$\frac{\partial \psi_f}{\partial t} = \frac{\partial \psi_f}{\partial p(t)} \frac{\partial p(t)}{\partial t} = \frac{\partial \psi_f}{\partial p(t)} \dot{p}(t), \tag{29}$$

$$\frac{\partial \psi_f}{\partial \tau} = \sum_\Phi \frac{\partial \psi_f}{\partial p_\phi(\tau)} \frac{\partial p_\phi(\tau)}{\partial \tau} = \sum_\Phi \frac{\partial \psi_f}{\partial p_\phi(\tau)} \dot{p}_\phi(\tau), \tag{30}$$

where the gradient of ψ_f to $p(t)$ and $p_\phi(\tau)$ is detailed in (5).

5) Swarm Reciprocal Avoidance J_r: We penalize the constraint points which are close to other agents' trajectories at global timestamp τ. Thus, the cost function of swarm reciprocal avoidance is defined as

$$J_r = \sum_\Phi \frac{T_i}{\kappa_i} \sum_{j=0}^{\kappa_i} \bar{\omega}_j \max\{\psi_{r_\phi}(p(t), \tau), 0\}^3, \tag{31}$$

$$\psi_{r_\phi}(p(t), \tau) = D_r^2 - d(p(t), p_\phi(\tau))^2, \tag{32}$$

$$d(p(t), p_\phi(\tau)) = \| \, p(t) - p_\phi(\tau) \, \|, \tag{33}$$

where D_r is the clearance between each agent.

The gradients of J_r w.r.t c_i and T_l are the same as (24) and (25), and $\partial \psi_{r_\phi} / \partial T_l$ is the same as (26). When $D_r^2 \geq d(p(t), p_\phi(\tau))^2$, the gradients of ψ_{r_ϕ} w.r.t c_i, t and τ are

$$\frac{\partial \psi_{r_\phi}}{\partial c_i} = -2\beta(t)(p(t) - p_\phi(\tau))^T, \tag{34}$$

$$\frac{\partial \psi_{r_\phi}}{\partial t} = -2(p(t) - p_\phi(\tau))^T \dot{p}(t), \tag{35}$$

$$\frac{\partial \psi_{r_\phi}}{\partial \tau} = 2(p(t) - p_\phi(\tau))^T \dot{p_\phi}(t). \tag{36}$$

6) Dynamical Feasibility J_d: We limit the maximum value of velocity, acceleration, and jerk to guarantee that the trajectory can be executed by the agent. Readers can refer to [15] for more details.

7) Uniform Distribution of Constraint Points J_u: The constraint points are expected to be space-uniform. Non-uniform constraint points may skip some small-sized obstacles, which could diminish the safety of the resulting trajectory. Therefore, the uniform distribution penalty J_u is optimized to prevent constraint points from gathering in certain locations. Readers can refer to [15] for more details.

3) 避障 J_o: 受到 [27], 避障惩罚的启发, J_o 使用欧几里得符号距离场（ESDF）进行计算。选择靠近障碍物的约束点的方法是

$$\psi_o(\hat{p}_{i,j}) = \begin{cases} d_{thr} - d(\hat{p}_{i,j}), & \text{if} \quad d(\hat{p}_{i,j}) < d_{thr}, \\ 0, & \text{if} \quad d(\hat{p}_{i,j}) \geq d_{thr}, \end{cases} \tag{17}$$

其中 d_{thr} 是安全阈值, $d(\hat{p}_{i,j})$ 是考虑点与其周围最近障碍物之间的距离。然后，通过计算采样约束函数的加权和来获得避障惩罚：

$$J_o = \frac{T_i}{\kappa_i} \sum_{j=0}^{\kappa_i} \bar{\omega}_j \max\{\psi_o(\hat{p}_{i,j}), 0\}^3, \tag{18}$$

其中 $(\bar{\omega}_0, \bar{\omega}_1, \cdots, \bar{\omega}_{\kappa_i - 1}, \bar{\omega}_{\kappa_i}) = (1/2, 1, \cdots, 1, 1$ 是遵循梯形规则 [28] 的正交系数。

J_o 关于 c_i 和 T_i 的梯度详细如下：

$$\frac{\partial J_o}{\partial c_i} = \frac{\partial J_o}{\partial \psi_o} \frac{\partial \psi_o}{\partial c_i}, \tag{19}$$

$$\frac{\partial J_o}{\partial T_i} = \frac{J_o}{T_i} + \frac{\partial J_o}{\partial \psi_o} \frac{\partial \psi_o}{\partial t} \frac{\partial t}{\partial T_i}, \tag{20}$$

$$\frac{\partial t}{\partial T_i} = \frac{j}{\kappa_i}, \qquad t = \frac{j}{\kappa_i} T_i, \tag{21}$$

其中 t 是片段上的相对时间。对于 $d(\hat{p}_{i,j}) < d_{thr}$ 的情况，梯度由以下公式给出：

$$\frac{\partial \psi_o}{\partial c_i} = -\beta(t) \nabla d^T, \qquad \frac{\partial \psi_o}{\partial t} = -\nabla d^T \dot{p}(t), \tag{22}$$

其中 ∇d 是在 $\hat{p}_{i,j}$ 中的 ESDF 梯度。否则，梯度变为 $\partial \psi_o / \partial c_i = \mathbf{0}$, $\partial \psi_o / \partial t = 0$。

4) 群体形成相似性 J_f: 在第III节中，我们设计了一种可微分的度量，用于量化群体形成之间的相似距离。在优化中，当前形成与目标形成之间的相似误差由 $\psi_f = f(p(t) \bigcup_\Phi p_\phi(\tau))$ 测量，其中 $f(\cdot)$ 在（4）中详细描述， Φ 代表其他智能体的集合。

由于 J_f 涉及其他智能体的轨迹，我们需要同时处理自身轨迹的相对时间 $t = jT_i/\kappa_i$ 和其他智能体轨迹的全局时间戳 $\tau = T_1 + ... + T_{i-1} + jT_i/\kappa_i$ 。 J_f 考虑任何 $1 \leq l \leq i$ 的前置时间 T_l ，其表达式为：

$$J_f = \frac{T_i}{\kappa_i} \sum_{j=0}^{\kappa_i} \bar{\omega}_j \max\{\psi_f(p(t), \bigcup_\Phi p_\phi(\tau)), 0\}^3. \tag{23}$$

关于 J_f 相对于 c_i 和 T_l 的梯度计算如下：

$$\frac{\partial J_f}{\partial c_i} = \frac{\partial J_f}{\partial \psi_f} \frac{\partial \psi_f}{\partial c_i}, \tag{24}$$

$$\frac{\partial J_f}{\partial T_l} = \frac{J_f}{T_l} + \frac{\partial J_f}{\partial \psi_f} \frac{\partial \psi_f}{\partial T_l}. \tag{25}$$

为了推导 $\partial \psi_f / \partial T_l$, ψ_f 需要对 t 和 τ 进行微分：

$$\frac{\partial \psi_f}{\partial T_l} = \frac{\partial \psi_f}{\partial t} \frac{\partial t}{\partial T_l} + \frac{\partial \psi_f}{\partial \tau} \frac{\partial \tau}{\partial T_l}, \tag{26}$$

$$\frac{\partial t}{\partial T_l} = \begin{cases} \frac{j}{\kappa_i}, & l = i, \\ 0, & l < i, \end{cases} \qquad \frac{\partial \tau}{\partial T_l} = \begin{cases} \frac{j}{\kappa_i}, & l = i, \\ 1, & l < i. \end{cases} \tag{27}$$

ψ_f 关于 c_i、 t 和 τ 的梯度由以下公式给出

$$\frac{\partial \psi_f}{\partial c_i} = \frac{\partial \psi_f}{\partial p(t)} \frac{\partial p(t)}{\partial c_i}, \tag{28}$$

$$\frac{\partial \psi_f}{\partial t} = \frac{\partial \psi_f}{\partial p(t)} \frac{\partial p(t)}{\partial t} = \frac{\partial \psi_f}{\partial p(t)} \dot{p}(t), \tag{29}$$

$$\frac{\partial \psi_f}{\partial \tau} = \sum_\Phi \frac{\partial \psi_f}{\partial p_\phi(\tau)} \frac{\partial p_\phi(\tau)}{\partial \tau} = \sum_\Phi \frac{\partial \psi_f}{\partial p_\phi(\tau)} \dot{p}_\phi(\tau), \tag{30}$$

其中， ψ_f 对 $p(t)$ 和 $p_\phi(\tau)$ 的梯度详见（5）。

5) 群体互避 J_r: 我们对在全局时间戳 τ 时接近其他智能体轨迹的约束点进行惩罚。因此，群体互避的代价函数定义为：

$$J_r = \sum_\Phi \frac{T_i}{\kappa_i} \sum_{j=0}^{\kappa_i} \bar{\omega}_j \max\{\psi_{r_\phi}(p(t), \tau), 0\}^3, \tag{31}$$

$$\psi_{r_\phi}(p(t), \tau) = D_r^2 - d(p(t), p_\phi(\tau))^2, \tag{32}$$

$$d(p(t), p_\phi(\tau)) = \| \, p(t) - p_\phi(\tau) \, \|, \tag{33}$$

其中 D_r 是每个智能体之间的间隙。

相对于 c_i 和 T_l 的 J_r 的梯度与（24）和（25）和（26）相同，并且与 $\partial \psi_{r_\phi} / \partial T_l$ 和 ψ_{r_ϕ} 相对于 c_i t τ 的梯度，以及，是 $D_r^2 \geq d(p(t), p_\phi(\tau))^2$, 当（ ）（ ）时，相对于的梯度，以及，是（ ）相同。

$$\frac{\partial \psi_{r_\phi}}{\partial c_i} = -2\beta(t)(p(t) - p_\phi(\tau))^T, \tag{34}$$

$$\frac{\partial \psi_{r_\phi}}{\partial t} = -2(p(t) - p_\phi(\tau))^T \dot{p}(t), \tag{35}$$

$$\frac{\partial \psi_{r_\phi}}{\partial \tau} = 2(p(t) - p_\phi(\tau))^T \dot{p_\phi}(t). \tag{36}$$

6) 动力学可行性 J_d: 我们限制速度、加速度和 jerk 的最大值，以保证轨迹可以由智能体执行。读者可以参考[15]获取更多细节。

7) 约束点的均匀分布 J_u: 约束点预计应具有空间上的均匀分布。非均匀的约束点可能会跳过一些小型障碍物，这可能会降低所生成轨迹的安全性。因此，均匀分布惩罚 J_u被优化以防止约束点在某些位置聚集。读者可以参考 [15]获取更多细节。

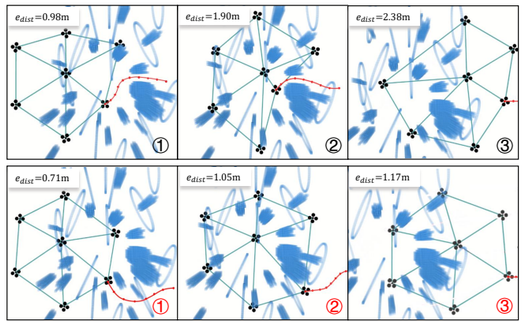


Fig. 3. Benchmark comparison between Turpin's method (the upper row) and our method (the lower row). With Turpin's method, robots could take non-colliding routes that actually impairs the formation. In contrast, with our method, robots make more reasonable decisions to hold the formation. Position error e_{dist} of each formation is shown in the figure.

TABLE I

FORMATION NAVIGATION METHODS COMPARISON

Scenario	Method	success rate(%)	e_{dist}(m)	e_{sim}
Sparse	Zhou's [23]	65	1.47	0.0162
	Turpin's [29]	85	1.08	0.0066
	Ours	**100**	**0.77**	**0.0032**
Medium	Zhou's [23]	15	-	-
	Turpin's [29]	75	1.55	0.0162
	Ours	**100**	**0.90**	**0.0045**
Dense	Zhou's [23]	0	-	-
	Turpin's [29]	60	2.01	0.0278
	Ours	**95**	**1.25**	**0.0107**

V. BENCHMARK

To demonstrate the efficiency and robustness of our method, benchmark comparisons are conducted with cutting-edge formation control methods. We compare our work with Zhou's method [23] and Turpin's method [29]. We implement the formation control method in Turpin's work and adapt it to the dense environments by adding our own obstacle avoidance strategy. However, unlike our work, in [23] and [29], changing the scale and rotation of the formation is not permitted during the flight. Hence, to evaluate the performance fairly, a new indicator of overall position error is proposed for formation flight.

Inspired by [30], in order to assess the overall position error e_{dist} between the current formation $\mathcal{F}^c$ and the desired one $\mathcal{F}^d$, we solve the following nonlinear optimization problem to find the best similarity transformation ($Sim(3)$ transformation) that aligns $\mathcal{F}^c$ with $\mathcal{F}^d$:

$$e_{dist} = \min_{\mathbf{R},\mathbf{t},s} \sum_{i=1}^{n} ||\mathbf{p}_i^d - (s\,\mathbf{R}\,\mathbf{p}_i^c + \mathbf{t})||^2. \tag{37}$$

Notations $\mathbf{p}_i^d$ and $\mathbf{p}_i^c$ represent the position of ith robot in formation $\mathcal{F}^d$ and $\mathcal{F}^c$, respectively. The $Sim(3)$ transformation is composed of a rotation $\mathbf{R} \in SO(3)$, a translation $\mathbf{t} \in \mathbb{R}^3$ and a scale expansion $s \in \mathbb{R}_+$. By optimizing

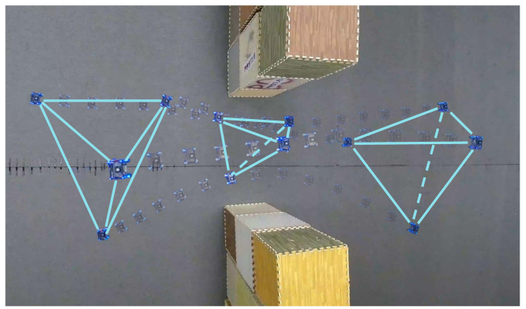


Fig. 4. Composite snapshots of a regular tetrahedron formation passing through a corridor. The swarm flies through the narrow space from right to left.

the transformation in (37) and applying to formations, the influence of scaling and rotation is squeezed out, so that all the formations can be equitably rated by measuring the position error w.r.t the desired formation. The larger the error e_{dist}, the more that $\mathcal{F}^c$ deviates from the desired shape $\mathcal{F}^d$. Besides the position error e_{dist}, methods are also compared over the success rate and the formation similarity error e_{sim} we proposed in Sec.III.

We simulate seven drones flying in a regular hexagon formation from one side of an obstacle-rich map to another side with a velocity limit of $0.5m/s$. The cluttered area is of 30×15m size, and three obstacle densities are tested for comparison. Parameters are finely tuned for the best performance of each compared method.

The result is summarized in Table I. It states that the success rate of Zhou's method [23] is unsatisfactory in the presence of dense obstacles. The multiple interacting potential fields used in their work tend to generate local minima near the corridors. Thus, drones are often trapped in deadlocks. Turpin's method [29] maintains formation by assigning desired relative positions to each pair of agents and then minimizing the relative error. However, these constraints are barely satisfied in cluttered scenarios, which raises the difficulty of finding feasible solutions and results in the lower success rates in Table I. Furthermore, Turpin's strategy focuses on the individual tracking error instead of the overall formation shape, which could result in unfavorable trajectories for formation flights, as shown in Fig.3.

In Table I, our method achieves better performance in terms of position error e_{dist} and formation similarity error e_{sim}. Moreover, our success rate is promising even with complex surroundings, since the scaling and rotational invariance of the proposed metric provides more flexibility for the planner to generate motion plans.

VI. REAL WORLD AND SIMULATION EXPERIMENTS

A. Implementation Details

Our method is integrated with an autonomous distributed aerial swarm system as shown in Fig.6. The swarm shares

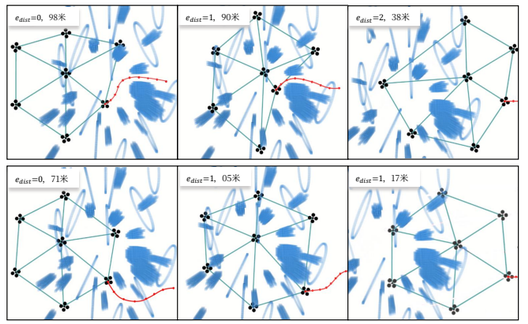


图3. Turpin方法（上排）与我们的方法（下排）的基准对比。采用Turpin的方法，机器人可以选择非碰撞路径，但实际上会削弱队形的完整性。相反，使用我们的方法，机器人会做出更合理的决策以保持队形。图中显示了每个队形的位置误差 e_{dist} 。

表1 队形导航方法比较

场景	方法	成功率（%）	e_{dist}(m)	e_{sim}
稀疏	周的 [23]	65	1.47	0.0162
	Turpin 的 [29]	85	1.08	0.0066
	我们的	**100**	**0.77**	**0.0032**
中等	周的 [23]	15	-	-
	Turpin的 [29]	75	1.55	0.0162
	我们的方法	**100**	**0.90**	**0.0045**
密集	周的 [23]	0	-	-
	Turpin的 [29]	60	2.01	0.0278
	我们的	**95**	**1.25**	**0.0107**

V. 基准

为了展示我们方法的效率和鲁棒性，进行了与尖端编队控制方法的基准比较。我们将我们的工作与Zhou的方法 [23] 和Turpin的方法 [29]进行了比较。我们实现了Turpin工作中的编队控制方法，并通过加入我们自己的避障策略将其适应于密集环境。然而，不同于我们的工作，在[23] 和 [29],中，飞行过程中不允许改变编队的规模和旋转。因此，为了公平评估性能，提出了一种新的整体位置误差指标，用于编队飞行。

受到 [30]，的启发，为了评估当前编队 $\mathcal{F}^c$ 与期望编队 $\mathcal{F}^d$ 之间的整体位置误差 e_{dist}，我们求解以下非线性优化问题，以找到最佳相似变换（$Sim(3)$ 变换），使得 $\mathcal{F}^c$ 与 $\mathcal{F}^d$ 对齐：

$$e_{dist} = \min_{\mathbf{R},\mathbf{t},s} \sum_{i=1}^{n} ||\mathbf{p}_i^d - (s\,\mathbf{R}\,\mathbf{p}_i^c + \mathbf{t})||^2. \tag{37}$$

符号 $\mathbf{p}_i^d$ 和 $\mathbf{p}_i^c$ 分别表示在编队 $\mathcal{F}^d$ 和 $\mathcal{F}^c$ 中第 i^{th} 机器人的位置。$Sim(3)$ 变换由一个旋转 $\mathbf{R} \in SO(3)$、一个平移 $\mathbf{t} \in \mathbb{R}^3$ 和一个比例扩展 $s \in \mathbb{R}_+$ 组成。通过优化

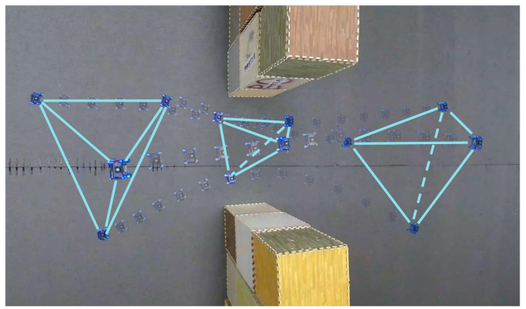


图4. 规则四面体队形穿过走廊的复合快照。群体从右向左穿过狭窄空间。

式（37）中的变换并应用于编队，缩小了比例和旋转的影响，从而所有编队都可以通过测量与期望编队的位置误差来公平地进行评估。误差越大e_{dist}，说明 $\mathcal{F}^c$ 偏离了期望的形状 $\mathcal{F}^d$。除了位置误差 e_{dist}之外，本文还将方法在成功率和我们在Sec.III中提出的编队相似度误差 e_{sim}方面进行了比较。

我们模拟七架无人机以规则六边形队形从一个充满障碍物的地图的一侧飞向另一侧，速度限制为$0.5m/s$。杂乱区域的大小为 30×15米，并测试了三种不同的障碍密度以进行比较。参数经过精细调优，以实现每种比较方法的最佳性能。

结果总结在表一中。结果显示，周氏方法 [23] 在密集障碍物存在的情况下成功率不理想。他们工作中使用的多重交互势场倾向于在走廊附近产生局部极小值。因此，无人机常常陷入死锁。Turpin的方法 [29]通过为每对智能体分配期望的相对位置并最小化相对误差来保持队形。然而，在杂乱场景中，这些约束几乎难以满足，这增加了找到可行解的难度，导致表一中的成功率较低。此外，Turpin的方法侧重于个体跟踪误差而非整体队形形状，这可能导致队形飞行的轨迹不理想，如图3所示。

在表I中，我们的方法在位置误差 e_{dist} 和编队相似性误差e_{sim}方面都取得了更好的性能。此外，即使在复杂环境中，我们的成功率也令人鼓舞，因为所提出的度量的缩放和旋转不变性为规划器生成运动规划提供了更大的灵活性。

六、实际世界与仿真实验

A. 实现细节

我们的方法集成于如图6所示的自主分布式空中蜂群系统中。蜂群通过广播网络共享

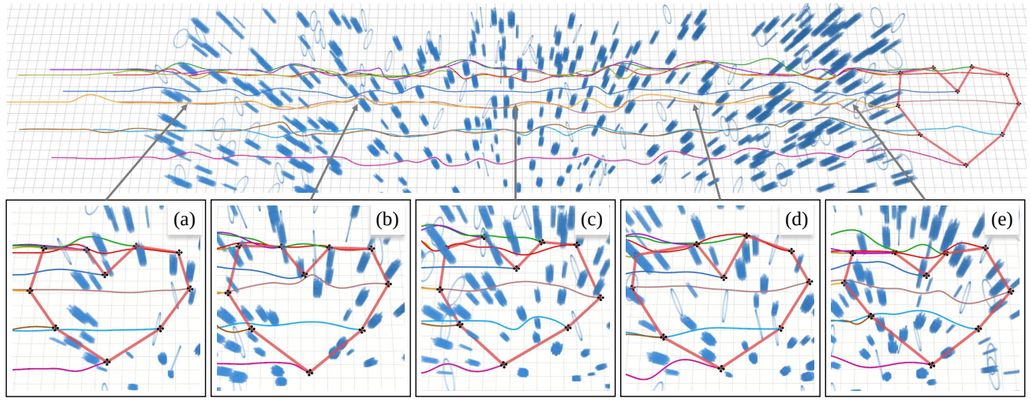


Fig. 5. A large-scale heart-shaped formation consisting of ten quadrotors traverses an unknown dense environment from the left side to right side. Snapshots (a)-(e) are uniformly sampled in the cluttered area. Colored curves show the executed trajectories of the formation flight.

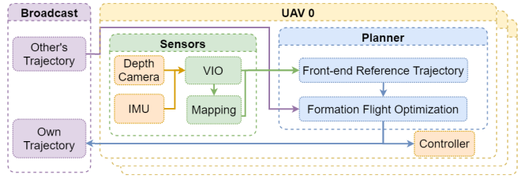


Fig. 6. System architecture of our distributed aerial swarm.

trajectories through a broadcast network, which is the only connection among all the quadrotors.

Each quadrotor is equipped with an Intel RealSense D435[3] stereo camera for imagery and depth sensing. In addition, software modules including state estimation, environment perception, trajectory planning, and flight control are all running with an onboard computer Xavier NX[4] in real-time.

In both real-world and simulation experiments, we generate the local trajectory by solving (13) per second and run collision check at a frequency of 100Hz. The unconstrained optimization problem is solved by an open-source library LBFGS-Lite[5]. All simulations are run on a desktop with an Intel i9-9900K CPU in real-time.

B. Real-world Experiments

Real-world experiments are designed to validate the feasibility and robustness of our method. In the first experiment, as shown in Fig.1, a 2-D hexagon formation consisting of seven quadrotors successfully traverses a obstacle-rich area without any collision. Twelve cylinder obstacles with a diameter of 0.3m are placed in the area. This test demonstrates that our method is able to maintain the formation for large-scale swarms in unknown complex environments.

[3]https://www.intelrealsense.com/depth-camera-d435/

[4]https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-xavier-nx/

[5]https://github.com/ZJU-FAST-Lab/LBFGS-Lite

In the second experiment, as shown in Fig.4, four quadrotors in a 3-D regular tetrahedron formation manage to pass through a narrow corridor safely. During the flight, the swarm adaptively rotates and compresses the formation shape in responding to the environmental changes. This test proves that the scaling and rotational invariance provides more flexibility for formation flights in constrained spaces.

C. Simulation Experiment

In order to testify the effectiveness of our method with large-scale irregular formations, we design a heart-shaped formation consisting of ten quadrotors. A cluttered area of $80 \times 20m$ with 300 cylinder obstacles and 80 circular obstacles is set up in simulation. As depicted in Fig.5, the swarm successfully avoids the obstacles and the desired formation is well preserved during the flight.

VII. Conclusions and Future Work

In this paper, we present a distributed swarm trajectory optimization method for formation flight in dense environments. A novel metric is proposed to measure the formation similarity, which is incorporated with a spatial-temporal optimization framework to generate swarm trajectories. The solid performance of our method in simulations and real-world experiments validates its practicality and efficiency.

In the future, we will be committed to further improving the robustness of our method. The challenges arise when the communication ranges of some robots are unpredictably narrowed. Also, in highly constrained environments, task reassignment among the robots could be necessary to resolve deadlocks timely. Finally, we hope to provide a complete solution of formation navigation in dense environments for the robotics community.

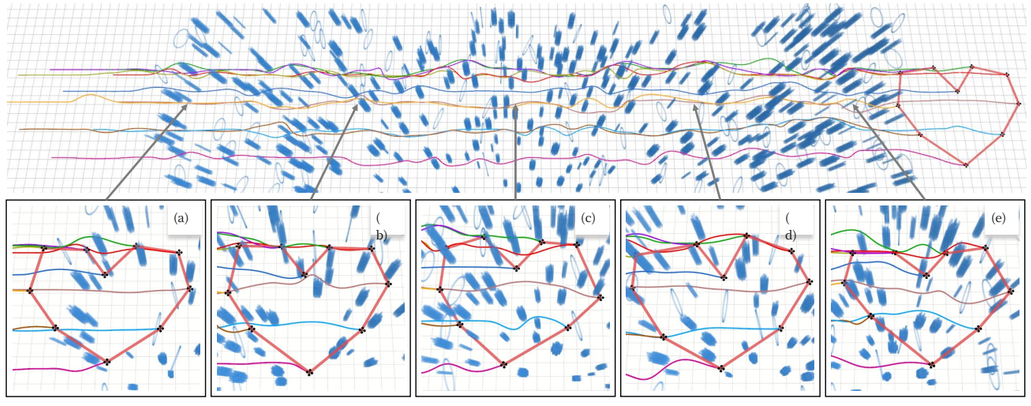


图 5. 大规模心形 由十个四旋翼组成的编队穿越一个未知的密集环境，从左侧到右侧。快照 (a)-(e) 以均匀采样的方式展开,在杂乱区域中编队飞行的轨迹。彩色曲线显示了编队飞行的执行轨迹。

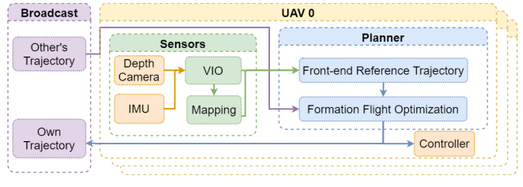


图6. 我们的分布式空中蜂群系统架构。

轨迹，这也是所有四旋翼之间唯一的连接方式。

每个四旋翼都配备了Intel RealSense D435[3]立体相机，用于图像采集和深度感知。此外，包括状态估计、环境感知、轨迹规划和飞行控制在内的软件模块都在机载计算机Xavier NX[4] 上实时运行。

在实际环境和仿真实验中，我们通过每秒求解一次（13）来生成局部轨迹，并以100Hz的频率进行碰撞检测。无约束优化问题由开源库LBFGS-Lite[5]求解。所有仿真均在配备Intel i9-9900K CPU的台式机上实时运行。

B. 实际环境实验

实际世界的实验旨在验证我们方法的可行性和鲁棒性。在第一个实验中，如图1所示，由七个四旋翼组成的二维六边形队形成功穿越了一个障碍丰富的区域，没有发生任何碰撞。该区域内放置了十二个直径为0.3米的圆柱形障碍物。此测试表明，我们的方法能够在未知复杂环境中维持大规模蜂群的队形。

[3]https://www.intelrealsense.com/depth-camera-d435/ [4] https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-xavier-nx/ [5] https://github.com/ZJU-FAST-Lab/LBFGS-Lite

在第二个实验中，如图4所示，四个四旋翼组成的三维正四面体队形成功穿过了一个狭窄的走廊。在飞行过程中，蜂群根据环境变化自适应地旋转和压缩队形形状。该测试证明了尺度和旋转不变性为受限空间中的队形飞行提供了更大的灵活性。

C. 仿真实验

为了验证我们方法在大规模不规则编队中的有效性，我们设计了一个由十架四旋翼组成的心形编队。在仿真中设置了一个杂乱区域，包含300个圆柱障碍物和80个圆形障碍物。如图5所示，群体成功避开了障碍物，并在飞行过程中很好地保持了预期的编队。

第七章 结论与未来工作

本文提出了一种用于密集环境中编队飞行的分布式群体轨迹优化方法。我们提出了一种新颖的度量指标，用于衡量编队的相似性，并将其融入空间-时间优化框架中，以生成群体轨迹。我们的方法在仿真和实际实验中的出色表现验证了其实用性和高效性。

未来，我们将致力于进一步提高我们方法的鲁棒性。当某些机器人之间的通信范围不可预测地缩小时，挑战就会出现。此外，在高度受限的环境中，可能需要在机器人之间重新分配任务，以及时解决死锁问题。最后，我们希望为机器人社区提供一种在密集环境中进行编队导航的完整解决方案。

References

[1] L. Marconi, C. Melchiorri, M. Beetz, D. Pangercic, R. Siegwart, S. Leutenegger, R. Carloni, S. Stramigioli, H. Bruyninckx, P. Doherty, A. Kleiner, V. Lippiello, A. Finzi, B. Siciliano, A. Sala, and N. Tomatis, "The sherpa project: Smart collaboration between humans and ground-aerial robots for improving rescuing activities in alpine environments," in *2012 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, 2012, pp. 1–4.

[2] N. Mahdoui, V. Frémont, and E. Natalizio, "Communicating multi-uav system for cooperative slam-based exploration," *Journal of Intelligent & Robotic Systems*, vol. 98, no. 2, pp. 325–343, 2020.

[3] K. Dorling, J. Heinrichs, G. G. Messier, and S. Magierowski, "Vehicle routing problems for drone delivery," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 47, no. 1, pp. 70–85, 2016.

[4] A. Jahn, R. J. Alitappeh, D. Saldaña, L. C. Pimenta, A. G. Santos, and M. F. Campos, "Distributed multi-robot coordination for dynamic perimeter surveillance in uncertain environments," in *2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2017, pp. 273–278.

[5] K.-K. Oh, M.-C. Park, and H.-S. Ahn, "A survey of multi-agent formation control," *Automatica*, vol. 53, pp. 424–440, 2015.

[6] J. Van Den Berg, S. J. Guy, M. Lin, and D. Manocha, "Reciprocal n-body collision avoidance," in *Robotics research*. Springer, 2011, pp. 3–19.

[7] J. Van Den Berg, J. Snape, S. J. Guy, and D. Manocha, "Reciprocal collision avoidance with acceleration-velocity obstacles," in *2011 IEEE International Conference on Robotics and Automation*. IEEE, 2011, pp. 3475–3482.

[8] D. Bareiss and J. Van den Berg, "Reciprocal collision avoidance for robots with linear dynamics using lqr-obstacles," in *2013 IEEE International Conference on Robotics and Automation*. IEEE, 2013, pp. 3847–3853.

[9] S. H. Arul and D. Manocha, "Dcad: Decentralized collision avoidance with dynamics constraints for agile quadrotor swarms," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 1191–1198, 2020.

[10] C. E. Luis and A. P. Schoellig, "Trajectory generation for multiagent point-to-point transitions via distributed model predictive control," *IEEE Robotics and Automation Letters*, vol. 4, no. 2, pp. 375–382, 2019.

[11] J. Park, J. Kim, I. Jang, and H. J. Kim, "Efficient multi-agent trajectory planning with feasibility guarantee using relative bernstein polynomial," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2020, pp. 434–440.

[12] D. Zhou, Z. Wang, S. Bandyopadhyay, and M. Schwager, "Fast, online collision avoidance for dynamic vehicles using buffered voronoi cells," *IEEE Robotics and Automation Letters*, vol. 2, no. 2, pp. 1047–1054, 2017.

[13] Y. Chen, M. Cutler, and J. P. How, "Decoupled multiagent path planning via incremental sequential convex programming," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2015, pp. 5954–5961.

[14] T. Baca, D. Hert, G. Loianno, M. Saska, and V. Kumar, "Model predictive trajectory tracking and collision avoidance for reliable outdoor deployment of unmanned aerial vehicles," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2018, pp. 6753–6760.

[15] X. Zhou, Z. Wang, X. Wen, J. Zhu, C. Xu, and F. Gao, "Decentralized spatial-temporal trajectory planning for multicopter swarms," *arXiv preprint arXiv:*, 2021.

[16] M. A. Lewis and K.-H. Tan, "High precision formation control of mobile robots using virtual structures," *Autonomous robots*, vol. 4, no. 4, pp. 387–403, 1997.

[17] M. C. De Gennaro and A. Jadbabaie, "Formation control for a cooperative multi-agent system using decentralized navigation functions," in *2006 American Control Conference*. IEEE, 2006, pp. 6–pp.

[18] T. Balch and R. C. Arkin, "Behavior-based formation control for multirobot teams," *IEEE transactions on robotics and automation*, vol. 14, no. 6, pp. 926–939, 1998.

[19] Z. Lin, W. Ding, G. Yan, C. Yu, and A. Giua, "Leader–follower formation via complex laplacian," *Automatica*, vol. 49, no. 6, pp. 1900–1906, 2013.

[20] Z. Han, L. Wang, and Z. Lin, "Local formation control strategies with undetermined and determined formation scales for co-leader vehicle networks," in *52nd IEEE Conference on Decision and Control*. IEEE, 2013, pp. 7339–7344.

[21] S. Zhao, "Affine formation maneuver control of multiagent systems," *IEEE Transactions on Automatic Control*, vol. 63, no. 12, pp. 4140–4155, 2018.

[22] J. Alonso-Mora, E. Montijano, M. Schwager, and D. Rus, "Distributed multi-robot formation control among obstacles: A geometric and optimization approach with consensus," in *2016 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2016, pp. 5356–5363.

[23] D. Zhou, Z. Wang, and M. Schwager, "Agile coordination and assistive collision avoidance for quadrotor swarms using virtual structures," *IEEE Transactions on Robotics*, vol. 34, no. 4, pp. 916–923, 2018.

[24] R. Van Parys and G. Pipeleers, "Distributed model predictive formation control with inter-vehicle collision avoidance," in *2017 11th Asian Control Conference (ASCC)*. IEEE, 2017, pp. 2399–2404.

[25] M. Tantardini, F. Ieva, L. Tajoli, and C. Piccardi, "Comparing methods for comparing networks," *Scientific Reports*, vol. 9, 11 2019.

[26] Z. Wang, X. Zhou, C. Xu, and F. Gao, "Geometrically constrained trajectory optimization for multicopters," *arXiv preprint arXiv:*, 2021.

[27] B. Zhou, F. Gao, L. Wang, C. Liu, and S. Shen, "Robust and efficient quadrotor trajectory generation for fast autonomous flight," *IEEE Robotics and Automation Letters*, vol. 4, no. 4, pp. 3529–3536, 2019.

[28] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes 3rd Edition: The Art of Scientific Computing*, 3rd ed. USA: Cambridge University Press, 2007.

[29] M. Turpin, N. Michael, and V. Kumar, "Trajectory design and control for aggressive formation flight with quadrotors," *Autonomous Robots*, vol. 33, no. 1, pp. 143–156, 2012.

[30] P. C. Lusk, X. Cai, S. Wadhwania, A. Paris, K. Fathian, and J. P. How, "A distributed pipeline for scalable, deconflicted formation flying," *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 5213–5220, 2020.

参考文献

[1] L. Marconi, C. Melchiorri, M. Beetz, D. Pangercic, R. Siegwart, S. Leutenegger, R. Carloni, S. Stramigioli, H. Bruyninckx, P. Doherty, A. Kleiner, V. Lippiello, A. Finzi, B. Siciliano, A. Sala 和 N. Tomatis, "The sherpa project: Smart collaboration between humans and ground-aerial robots for improving rescuing activities in alpine environments," 载于 2012 年 IEEE 国际安全、安保与救援机器人研讨会（SSRR），2012 年，第 1–4 页。[2] N. Mahdoui, V. Fr´emont 和 E. Natalizio, "用于合作 SLAM 探索的多无人机系统通信，"《智能与机器人系统杂志》，第 98 卷，第 2 期，页 325–343，2020 年。[3] K. Dorling, J. Heinrichs, G. G. Messier 和 S. Magierowski, "无人机配送的车辆路径问题，"《IEEE 系统、人类与控制论：系统》杂志，第 47 卷，第 1 期，页 70–85，2016 年。[4] A. Jahn, R. J. Alitappeh, D. Salda˜na, L. C. Pimenta, A. G. Santos 和 M. F. Campos, "在不确定环境中进行动态周界监控的分布式多机器人协调，" 载于 2017 年 IEEE 机器人与自动化国际会议（ICRA）。IEEE，2017 年，第 273–278 页。[5] K.-K. Oh, M.-C. Park 和 H.-S. Ahn, "多智能体编队控制综述，"《自动控制》杂志，第 53 卷，页 424–440，2015 年。[6] J. Van Den Berg, S. J. Guy, M. Lin 和 D. Manocha, "互惠 n 体碰撞避免，" 载于机器人研究。Springer，2011 年，第 3–19 页。[7] J. Van Den Berg, J. Snape, S. J. Guy 和 D. Manocha, "基于加速度-速度障碍的互惠碰撞避免，" 载于 2011 年 IEEE 机器人与自动化国际会议。IEEE，2011 年，第 3475–3482 页。[8] D. Bareiss 和 J. Van den Berg, "使用 LQR 障碍的线性动力学机器人互惠碰撞避免，" 载于 2013 年 IEEE 机器人与自动化国际会议。IEEE，2013 年，第 3847–3853 页。[9] S. H. Arul 和 D. Manocha, "Dcad：具有动力学约束的去中心化碰撞避免，用于敏捷四旋翼群，"《IEEE 机器人与自动化快报》，第 5 卷，第 2 期，页 1191–1198，2020 年。[10] C. E. Luis 和 A. P. Schoellig, "通过分布式模型预测控制实现多智能体点对点轨迹生成，"《IEEE 机器人与自动化快报》，第 4 卷，382，2019 年，第 2 页，页 375。[11] J. Park, J. Kim, I. Jang 和 H. J. Kim, "使用相对伯恩斯坦多项式的高效多智能体轨迹规划与可行性保证，" 载于 2020 年 IEEE 机器人与自动化国际会议（ICRA）。IEEE，2020 年，第 434–440 页。[12] D. Zhou, Z. Wang, S. Bandyopadhyay 和 M. Schwager, "使用缓冲 Voronoi 单元的动态车辆快速在线碰撞避免，"《IEEE 机器人与自动化快报》，第 2 卷，第 2 期，页 1047–1054，2017 年。[13] Y. Chen, M. Cutler 和 J. P. How, "通过增量序贯凸规划实现的解耦多智能体路径规划，" 载于 2015 年 IEEE 机器人与自动化国际会议（ICRA）。IEEE，2015 年，第 5954–5961 页。[14] T. Baca, D. Hert, G. Loianno, M. Saska 和 V. Kumar, "用于可靠户外部署无人机的模型预测轨迹跟踪与碰撞避免，" 载于 2018 年 IEEE/RSJ 国际智能机器人与系统会议（IROS）。IEEE，2018 年，第 6753–6760 页。[15] X. Zhou, Z. Wang, X. Wen, J. Zhu, C. Xu 和 F. Gao, "多旋翼群的去中心化时空轨迹规划，" arXiv 预印本 arXiv:，2021 年。[16] M. A. Lewis 和 K.-H. Tan, "使用虚拟结构实现移动机器人高精度编队控制，"《自主机器人》, 第 4 卷，第 4 期，页 387–403，1997 年。[17] M. C. De Gennaro 和 A. Jadbabaie, "使用去中心化导航函数的合作多智能体系统的编队控制，" 载于 2006 年美国控制会议。IEEE，2006 年，第 6 页。[18] T. Balch 和 R. C. Arkin, "基于行为的多机器人队伍编队控制，"《IEEE 机器人与自动化学报》，第 14 卷，第 6 期，页 926–939，1998 年。[19] Z. Lin, W. Ding, G. Yan, C. Yu 和 A. Giua, "通过复合拉普拉斯实现的领袖-追随者编队，"《自动控制》, 第 49 卷，第 6 期，页 1900–1906，2013 年。[20] Z. Han, L. Wang 和 Z. Lin, "具有未确定和已确定编队规模的局部编队控制策略，用于共同领导车辆网络，" 载于第 52 届 IEEE 决策与控制会议。IEEE，2013 年，第 7339–7344 页。

[21] S. Zhao, "多智能体系统的仿射编队机动控制"，*IEEE*自动控制汇刊，第63卷，第12期，第4140–4155页，2018年。[22] J. Alonso-Mora，E. Montijano，M. Schwager，和D. Rus，"障碍物间的分布式多机器人编队控制：一种几何与优化方法与共识"，收录于*2016 IEEE*机器人与自动化国际会议（*ICRA*）。IEEE，2016年，第5356–5363页。[23] D. Zhou，Z. Wang，和M. Schwager，"利用虚拟结构实现四旋翼群的敏捷协调与辅助避碰"，*IEEE*机器人汇刊，第34卷，第4期，第916–923页，2018年。[24] R. Van Parys 和 G. Pipeleers，"具有车间碰撞避免的分布式模型预测编队控制"，收录于*2017*年第*11*届亚洲控制会议（*ASCC*）。IEEE，2017年，第2399–2404页。[25] M. Tantardini，F. Ieva，L. Tajoli 和 C. Piccardi，"网络比较方法的比较"，科学报告，第9卷，2019年11月。[26] Z. Wang，X. Zhou，C. Xu 和 F. Gao，"多旋翼机的几何约束轨迹优化"，*arXiv*预印本 *arXiv:*，2021年。[27] B. Zhou，F. Gao，L. Wang，C. Liu 和 S. Shen，"用于快速自主飞行的鲁棒高效四旋翼轨迹生成"，*IEEE*机器人与自动化快报，第4卷，第4期，第3529–3536页，2019年。[28] W. H. Press，S. A. Teukolsky，W. T. Vetterling 和 B. P. Flannery，《数值算法第三版：科学计算的艺术》，第3版，美国：剑桥大学出版社，2007年。[29] M. Turpin，N. Michael 和 V. Kumar，"用于激进编队飞行的轨迹设计与控制"，自主机器人，第33卷，第1期，第143–156页，2012年。[30] P. C. Lusk，X. Cai，S. Wadhwania，A. Paris，K. Fathian 和 J. P. How，"一种用于可扩展、无冲突编队飞行的分布式管道"，*IEEE*机器人与自动化快报，第5卷，编号5220，2020年，第5213页。